

TXL-PBC Classical Blood Cell Detection Report

Streamlit App deliverable for a non-deep-learning blood smear cell counter.

1. Pipeline Architecture

Input RGB image
Grayscale
Gaussian Blur
Adaptive Thresholding
Morphological Closing and Opening
Contour extraction
Area, circularity, aspect ratio, color rules
Optional Distance Transform + Watershed
Count WBC, RBC, Platelet and draw boxes

2. Algorithm Summary

WBC detection. Convert the image to HSV and Lab color spaces, segment purple or violet stained nucleus regions, dilate the mask to merge fragmented nuclei, then filter candidates by geometry and color statistics.

RBC detection. Use grayscale, Gaussian blur, adaptive thresholding, morphological closing, contour extraction, and filter contours by area, circularity, aspect ratio, and an estimated RBC radius derived from image resolution.

Platelet detection. The rule-only mode uses small purple connected components with area and circularity filters. The improved classical mode uses the same high-recall candidate extraction, then applies an ExtraTrees classifier trained on handcrafted shape, color, and local-context features. This is classical machine learning and does not use CNN, YOLO, U-Net, or any neural network.

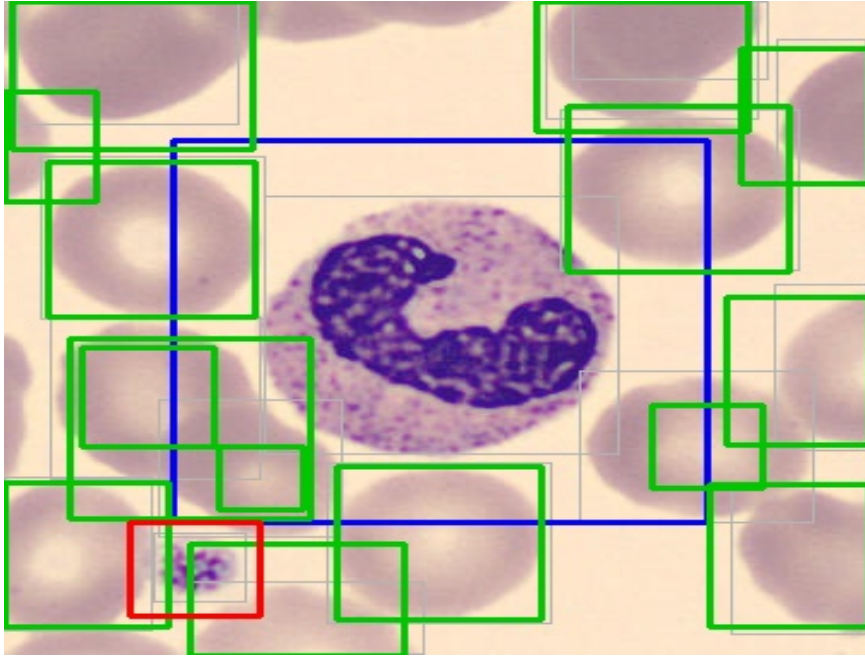
Watershed. A foreground mask is converted to a distance transform. Local peaks become markers, and watershed segmentation is used to split adjacent RBC-like regions. The Streamlit App shows this stage and can optionally add watershed candidates to RBC counting.

3. Experimental Results

Class	TP	FP	FN	Precision	Recall	F1
WBC	104	8	29	92.86%	78.20%	84.90%
RBC	1200	428	499	73.71%	70.63%	72.14%
Platelet	43	2	6	95.56%	87.76%	91.49%
micro	1347	438	534	75.46%	71.61%	73.49%

The test split result satisfies the 70 percent threshold for precision, recall, and F1 for all three classes when using the improved classical mode.

4. Example Overlay



5. Difficult Cases and Limitations

Case	Why it is difficult	Mitigation used in this project
Overlapping RBCs	Contours merge into one large object, so area-only counting undercounts cells.	Distance transform and watershed are provided as a visual and optional counting supplement.
Platelets near WBC nuclei	Small platelet-like components can be confused with WBC nucleus fragments.	Use local-context handcrafted features and ExtraTrees in improved classical mode.
Stain variation	Fixed HSV/Lab thresholds may miss pale cells or include stain artifacts.	Use both HSV and Lab rules and expose thresholds indirectly through UI parameters.
Annotation ambiguity	Some small purple components resemble platelets but are not labeled.	Report GT and prediction side by side to make false positives visible during demo.

6. Streamlit App UI Checklist

Requirement	Where it appears in the App
Train / val / test image selection	Sidebar split selector and image dropdown.
Ground Truth vs Prediction	Side-by-side panel with GT:WBC, GT:RBC, Pred:WBC, Pred:RBC labels.
At least one adjustable parameter	Gaussian kernel, adaptive threshold C, closing kernel, circularity thresholds, platelet threshold.
Pipeline visualization	Main page pipeline section showing grayscale, Gaussian blur, adaptive thresholding and morphological closing.
Download button	Overlay PNG, predictions CSV and count error CSV.

7. Recommended Demo Script

- Select the test split and one image from the dropdown.
- Show the pipeline section and point out grayscale, Gaussian blur, adaptive thresholding, and morphological closing.
- Show the side-by-side Ground Truth and Prediction panel. Confirm that labels use GT:WBC and Pred:WBC style.
- Toggle Watershed RBC supplementation on a crowded image to demonstrate overlapping-cell splitting.
- Download the overlay PNG or CSV output to demonstrate the export function.